

DERIVATION OF DC-LINK CAPACITOR AND DOUBLE-FREQUENCY RIPPLE POWER

The DC-bus voltage is composed of a steady-state component V_0 and a second-harmonic (double-frequency) ripple component, given as

$$V_g^{\text{DC}} = V_g^{\text{DC},0} + \underbrace{V_g^{\text{DC},2\omega} \sin(2\omega t + \theta_{v,g}^{\text{DC}})}_{\text{double-frequency ripple}}. \quad (1)$$

where V_g^{DC} is the instantaneous DC-bus voltage of converter g , $V_g^{\text{DC},0}$ is the steady-state (average) component of the DC-bus voltage, $V_g^{\text{DC},2\omega}$ is the amplitude of the double-frequency voltage ripple, ω is the fundamental grid angular frequency, and $\theta_{v,g}^{\text{DC}}$ is the phase angle of the double-frequency voltage ripple.

Similarly, the DC-bus current is composed of an average component and a double-frequency ripple component:

$$I_g^{\text{DC}} = I_g^{\text{DC},0} + \underbrace{I_g^{\text{DC},2\omega} \sin(2\omega t + \theta_{i,g}^{\text{DC}})}_{\text{double-frequency ripple}}. \quad (2)$$

where I_g^{DC} is the instantaneous DC-bus current of converter g , $I_g^{\text{DC},0}$ is its average (DC) component, $I_g^{\text{DC},2\omega}$ is the amplitude of its double-frequency ripple component, and $\theta_{i,g}^{\text{DC}}$ is the corresponding current-ripple phase angle.

The voltage- and current-ripple phase angles are written separately as $\theta_{v,g}^{\text{DC}}$ and $\theta_{i,g}^{\text{DC}}$, respectively, to allow for a phase displacement between the double-frequency voltage and current components.

DC-Bus Power and Double-Frequency Component

The instantaneous DC-bus power is obtained from the product of (1) and (2):

$$\begin{aligned} \mathbf{P}_g^{\text{DC}} &= V_g^{\text{DC}} I_g^{\text{DC}} \\ &= [V_g^{\text{DC},0} + V_g^{\text{DC},2\omega} \sin(2\omega t + \theta_{v,g}^{\text{DC}})] \\ &\quad \times [I_g^{\text{DC},0} + I_g^{\text{DC},2\omega} \sin(2\omega t + \theta_{i,g}^{\text{DC}})] \\ &= \underbrace{V_g^{\text{DC},0} I_g^{\text{DC},0}}_{\text{average-power contribution}} \\ &\quad + \underbrace{V_g^{\text{DC},0} I_g^{\text{DC},2\omega} \sin(2\omega t + \theta_{i,g}^{\text{DC}})}_{\text{double-frequency contribution}} \\ &\quad + \underbrace{V_g^{\text{DC},2\omega} I_g^{\text{DC},0} \sin(2\omega t + \theta_{v,g}^{\text{DC}})}_{\text{double-frequency contribution}} \\ &\quad + \underbrace{V_g^{\text{DC},2\omega} I_g^{\text{DC},2\omega} \sin(2\omega t + \theta_{v,g}^{\text{DC}}) \sin(2\omega t + \theta_{i,g}^{\text{DC}})}_{\text{ripple-ripple coupling term}} \end{aligned} \quad (3)$$

Equation (3) shows that the instantaneous DC-bus power contains contributions from the average voltage and current components, two double-frequency cross terms, and a ripple-ripple coupling term. The product of the two 2ω ripple components produces DC and 4ω terms and does not directly constitute the dominant 2ω component.

Therefore, isolating the terms oscillating at 2ω , the double-frequency component of the DC-bus power is

$$\begin{aligned} \mathbf{P}_g^{\text{DC},2\omega}(t) &= \underbrace{V_g^{\text{DC},0} I_g^{\text{DC},2\omega} \sin(2\omega t + \theta_{i,g}^{\text{DC}})}_{\text{average voltage} \times \text{current ripple}} \\ &\quad + \underbrace{V_g^{\text{DC},2\omega} I_g^{\text{DC},0} \sin(2\omega t + \theta_{v,g}^{\text{DC}})}_{\text{voltage ripple} \times \text{average current}}. \end{aligned} \quad (4)$$

Equation (4) represents the double-frequency power oscillation present at the DC bus. This oscillatory power must be accommodated by the DC-link energy-storage element to limit the resulting DC-bus voltage ripple.

DC-Link Capacitor Current

To relate the double-frequency DC-bus power to the required DC-link capacitance, the current through the DC-link capacitor is distinguished from the total DC-bus current in (2). The capacitor current is denoted by $I_{C,g}^{\text{DC}}$ and is given by

$$I_{C,g}^{\text{DC}} = C_g \frac{dV_g^{\text{DC}}}{dt}. \quad (5)$$

Substituting (1) into (5) yields

$$\begin{aligned} I_{C,g}^{\text{DC}} &= C_g \frac{d}{dt} [V_g^{\text{DC},0} + V_g^{\text{DC},2\omega} \sin(2\omega t + \theta_{v,g}^{\text{DC}})] \\ &= 2\omega C_g V_g^{\text{DC},2\omega} \cos(2\omega t + \theta_{v,g}^{\text{DC}}). \end{aligned} \quad (6)$$

Hence, the amplitude of the double-frequency current through the DC-link capacitor is

$$I_{C,g}^{\text{DC},2\omega} = 2\omega C_g V_g^{\text{DC},2\omega}. \quad (7)$$

It is important to distinguish I_g^{DC} from $I_{C,g}^{\text{DC}}$: I_g^{DC} denotes the total instantaneous DC-bus current used in the DC-bus power expression, whereas $I_{C,g}^{\text{DC}}$ denotes only the current flowing through the DC-link capacitor. Consequently, $I_g^{\text{DC},2\omega}$ in (2) denotes the double-frequency component of the DC-bus current, while $I_{C,g}^{\text{DC},2\omega}$ in (7) denotes the amplitude of the capacitor ripple current.

Relationship Between DC-Bus Ripple Power and DC-Link Capacitance

The double-frequency power oscillation at the DC bus causes a periodic exchange of energy with the DC-link capacitor. Accordingly, the double-frequency power buffered by the capacitor can be obtained from the instantaneous capacitor power

$$\mathbf{P}_{C,g}^{\text{DC}} = V_g^{\text{DC}} I_{C,g}^{\text{DC}}. \quad (8)$$

Substituting (1) and (6) into (8) gives

$$\begin{aligned} \mathbf{P}_{C,g}^{\text{DC}} &= [V_g^{\text{DC},0} + V_g^{\text{DC},2\omega} \sin(2\omega t + \theta_{v,g}^{\text{DC}})] \\ &\quad \times [2\omega C_g V_g^{\text{DC},2\omega} \cos(2\omega t + \theta_{v,g}^{\text{DC}})]. \end{aligned} \quad (9)$$

Expanding the expression gives

$$\begin{aligned} \mathbf{P}_{C,g}^{\text{DC}} &= \underbrace{2\omega C_g V_g^{\text{DC},0} V_g^{\text{DC},2\omega} \cos(2\omega t + \theta_{v,g}^{\text{DC}})}_{\text{dominant double-frequency power}} \\ &\quad + \underbrace{2\omega C_g (V_g^{\text{DC},2\omega})^2 \sin(2\omega t + \theta_{v,g}^{\text{DC}}) \cos(2\omega t + \theta_{v,g}^{\text{DC}})}_{\text{higher-order ripple contribution}}. \end{aligned}$$

Applying the trigonometric identity

$$\sin(\alpha) \cos(\alpha) = \frac{1}{2} \sin(2\alpha),$$

(10) becomes

$$\begin{aligned} \mathbf{P}_{C,g}^{\text{DC}} = & \underbrace{2\omega C_g V_g^{\text{DC},0} V_g^{\text{DC},2\omega} \cos(2\omega t + \theta_{v,g}^{\text{DC}})}_{\text{dominant } 2\omega \text{ component}} \\ & + \underbrace{\omega C_g (V_g^{\text{DC},2\omega})^2 \sin(4\omega t + 2\theta_{v,g}^{\text{DC}})}_{\text{higher-order } 4\omega \text{ component}}. \end{aligned} \quad (11)$$

Since

$$V_g^{\text{DC},2\omega} \ll V_g^{\text{DC},0},$$

the second term in (11) is of order $\mathcal{O}((V_g^{\text{DC},2\omega})^2)$ and is neglected under the small-ripple approximation. Thus, only the dominant 2ω component is retained:

$$\mathbf{P}_{C,g}^{\text{DC},2\omega}(t) \approx 2\omega C_g V_g^{\text{DC},0} V_g^{\text{DC},2\omega} \cos(2\omega t + \theta_{v,g}^{\text{DC}}). \quad (12)$$

Under the assumption that the dominant double-frequency DC-bus power oscillation identified in (4) is buffered by the DC-link capacitor, its amplitude is represented by $\mathbf{P}_g^{2\omega}$. Therefore,

$$\begin{aligned} \mathbf{P}_g^{2\omega} &= V_g^{\text{DC},0} I_{C,g}^{\text{DC},2\omega} \\ &= 2\omega C_g V_g^{\text{DC},0} V_g^{\text{DC},2\omega}. \end{aligned} \quad (13)$$

Equation (13) establishes the relationship between the double-frequency DC-bus power that must be buffered, the DC-link capacitance, and the corresponding DC-bus voltage-ripple amplitude.

DC-LINK CAPACITOR AND RIPPLE POWER COST MODEL

The DC-link capacitor cost follows the analytical expression in [1], where the cost of a metallized polypropylene film capacitor is expressed as a function of the rated voltage V_0 and capacitance C_g :

$$C_{\text{dc}} = a_{\text{film}} + b_{\text{film}} V_0 + c_{\text{film}} C_g, \quad (14)$$

with $a_{\text{film}} = -1.19574$ \$, $b_{\text{film}} = 0.06429852$ \$/V, and $c_{\text{film}} = 0.00283842$ \$/ μF .

Using $V_0 = 1100$ V and $C_g = 40$ μF , the reference capacitor cost is

$$C_{\text{dc}} \approx \$69.72.$$

For the linear cost representation adopted in the sizing optimization, the reference average cost per unit capacitance is obtained from this design point as

$$C_{c_g} = \frac{C_{\text{dc}}}{C_g} = \frac{69.72}{40} \approx 1.743 \text{ \$}/\mu\text{F} \approx 1.743 \times 10^6 \text{ \$}/\text{F}. \quad (15)$$

Accordingly, the capacitor cost used in the sizing optimization is represented by the adopted linear approximation

$$\hat{C}_{\text{dc}}(C_g) = C_{c_g} C_g, \quad (16)$$

where C_{c_g} represents the reference average unit capacitance cost obtained from the selected capacitor design point. Thus, C_{c_g} should not be interpreted as the marginal capacitance cost coefficient of the original analytical model in (14).

A. Relationship between the Capacitance and Ripple Power

From (13), the required capacitance C_g can be expressed in terms of the double-frequency ripple-power amplitude as

$$C_g = \frac{\mathbf{P}_g^{2\omega}}{2\omega V_g^{\text{DC},0} V_g^{\text{DC},2\omega}}. \quad (17)$$

The steady-state and double-frequency components of the DC-bus voltage can be related through the peak-to-peak voltage ripple. From (1), the maximum DC-bus voltage occurs when $\sin(2\omega t + \theta_{v,g}^{\text{DC}}) = 1$, giving

$$V_g^{\text{DC},\text{max}} = V_g^{\text{DC},0} + V_g^{\text{DC},2\omega}. \quad (18)$$

Similarly, the minimum DC-bus voltage occurs when $\sin(2\omega t + \theta_{v,g}^{\text{DC}}) = -1$, giving

$$V_g^{\text{DC},\text{min}} = V_g^{\text{DC},0} - V_g^{\text{DC},2\omega}. \quad (19)$$

Hence, the peak-to-peak DC-bus voltage ripple is

$$\begin{aligned} \Delta V_g^{\text{DC},pp} &= V_g^{\text{DC},\text{max}} - V_g^{\text{DC},\text{min}} \\ &= (V_g^{\text{DC},0} + V_g^{\text{DC},2\omega}) - (V_g^{\text{DC},0} - V_g^{\text{DC},2\omega}) \\ &= 2V_g^{\text{DC},2\omega}. \end{aligned} \quad (20)$$

According to [2] and related literature, the maximum allowable peak-to-peak DC-bus voltage ripple is constrained as

$$\Delta V_g^{\text{DC},pp} \leq 0.10 V_g^{\text{DC},0}. \quad (21)$$

For the capacitor-sizing calculation, the ripple constraint is assumed to be active at the maximum allowable value. Therefore,

$$\Delta V_g^{\text{DC},pp} = 0.10 V_g^{\text{DC},0}. \quad (22)$$

Substituting (20) into (22) yields

$$2V_g^{\text{DC},2\omega} = 0.10 V_g^{\text{DC},0}. \quad (23)$$

Hence, the relationship between $V_g^{\text{DC},0}$ and $V_g^{\text{DC},2\omega}$ becomes

$$V_g^{\text{DC},2\omega} = 0.05 V_g^{\text{DC},0}. \quad (24)$$

Substituting (24) into (17) gives

$$C_g = \frac{\mathbf{P}_g^{2\omega}}{0.1\omega (V_g^{\text{DC},0})^2}. \quad (25)$$

Equation (25) therefore relates the required DC-link capacitance to the double-frequency power that must be buffered for the adopted DC-bus voltage and maximum allowable voltage-ripple limit.

B. Ripple Power Cost Model

The DC-link ripple-power cost is obtained by relating the capacitor cost to the double-frequency ripple power through the required DC-link capacitance. Specifically, the adopted capacitor-cost approximation is a function of C_g , while C_g is itself a function of the double-frequency ripple power $\mathbf{P}_g^{2\omega}$. Therefore, the functional dependence can be expressed as

$$\hat{C}_{\text{dc}} = \hat{C}_{\text{dc}}(C_g(\mathbf{P}_g^{2\omega})). \quad (26)$$

From (25), the required DC-link capacitance is linearly related to the double-frequency ripple power as

$$C_g = \frac{\mathbf{P}_g^{2\omega}}{0.1\omega \left(V_g^{\text{DC},0}\right)^2}. \quad (27)$$

Since this relationship is linear in $\mathbf{P}_g^{2\omega}$, the required capacitance per unit of double-frequency ripple power is

$$\frac{dC_g}{d\mathbf{P}_g^{2\omega}} = \frac{1}{0.1\omega \left(V_g^{\text{DC},0}\right)^2}. \quad (28)$$

The quantity $dC_g/d\mathbf{P}_g^{2\omega}$ has units of F/W and represents the required increase in DC-link capacitance per unit increase in double-frequency ripple power under the adopted DC-bus voltage-ripple limit.

Similarly, for the linear capacitor-cost approximation adopted in the sizing optimization,

$$\hat{C}_{\text{dc}}(C_g) = C_{c_g} C_g, \quad (29)$$

where C_{c_g} is the reference average cost per unit capacitance obtained from (15). Since (29) is linear in C_g , differentiating the adopted cost approximation with respect to C_g gives

$$\frac{d\hat{C}_{\text{dc}}}{dC_g} = C_{c_g}. \quad (30)$$

Thus, within the adopted linear cost approximation, $d\hat{C}_{\text{dc}}/dC_g$ represents the DC-link capacitor cost per unit increase in capacitance and has units of \$/F.

Since $\hat{C}_{\text{dc}}$ is a function of C_g , and C_g is in turn a function of $\mathbf{P}_g^{2\omega}$, the DC-link cost per unit of double-frequency ripple power follows directly from the chain rule:

$$\begin{aligned} \frac{d\hat{C}_{\text{dc}}}{d\mathbf{P}_g^{2\omega}} &= \frac{d\hat{C}_{\text{dc}}}{dC_g} \frac{dC_g}{d\mathbf{P}_g^{2\omega}} \\ &= \underbrace{C_{c_g}}_{\text{cost per unit capacitance}} \underbrace{\frac{1}{0.1\omega \left(V_g^{\text{DC},0}\right)^2}}_{\text{capacitance per unit ripple power}} \\ &= \frac{C_{c_g}}{0.1\omega \left(V_g^{\text{DC},0}\right)^2}. \end{aligned} \quad (31)$$

Accordingly, the DC-link ripple-power cost coefficient is defined as

$$c_{2\omega} = \frac{d\hat{C}_{\text{dc}}}{d\mathbf{P}_g^{2\omega}} = \frac{C_{c_g}}{0.1\omega \left(V_g^{\text{DC},0}\right)^2}. \quad (32)$$

The dimensional consistency of the chain-rule relationship can be verified as

$$\underbrace{\left[\frac{d\hat{C}_{\text{dc}}}{dC_g}\right]}_{\$/\text{F}} \underbrace{\left[\frac{dC_g}{d\mathbf{P}_g^{2\omega}}\right]}_{\text{F}/\text{W}} = \underbrace{\left[\frac{d\hat{C}_{\text{dc}}}{d\mathbf{P}_g^{2\omega}}\right]}_{\$/\text{W}}. \quad (33)$$

Using

$$\omega = 2\pi f, \quad f = 60 \text{ Hz}, \quad V_g^{\text{DC},0} = V_0 = 1100 \text{ V},$$

$$C_{c_g} = 1.743 \times 10^6 \text{ $/F},$$

the DC-link ripple-power cost coefficient becomes

$$\begin{aligned} c_{2\omega} &= \frac{1.743 \times 10^6}{0.1(2\pi \times 60)(1100)^2} \\ &\approx 0.0382 \text{ $/W} = 38.2 \text{ $/kW}. \end{aligned} \quad (34)$$

Thus, $c_{2\omega} = 38.2 \text{ $/kW}$ represents the adopted linear DC-link cost per unit of double-frequency ripple power. The coefficient is obtained through the functional dependence of the capacitor cost on the required capacitance and, in turn, of the required capacitance on the double-frequency ripple power.

REFERENCES

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